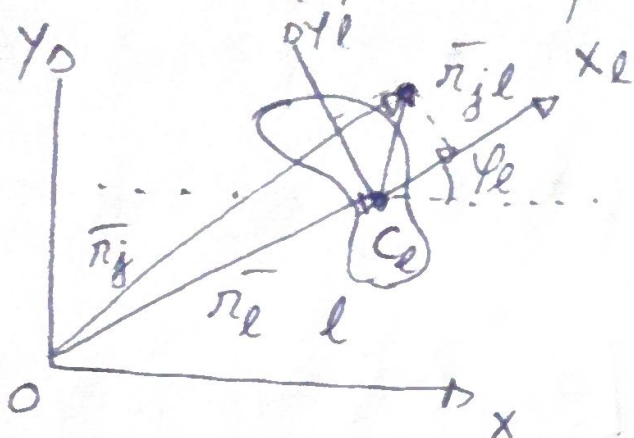


Considerăm un sistem de n corpuri, fiecare fiind compus de un număr arbitrar de puncte aflate în mișcare plan-paralelă.



Fiecare corp are un sistem de coordonate x_e, y_e , aflat în centrul său de masă. Sistemul este considerat aflat la un unghi φ_e ce corespunde mișcării corpului. Versorii sistemului sunt $\bar{i}_e, \bar{j}_e$ necomstanți.

$$\bar{r}_{je} = d_{xje} \bar{i}_e + d_{yje} \bar{j}_e$$

$$\bar{i}_e = \cos \varphi_e \bar{i} + \sin \varphi_e \bar{j}$$

$$\bar{j}_e = \sin \varphi_e \bar{i} + \cos \varphi_e \bar{j}$$

$$\Rightarrow \bar{r}_{je} = (d_{xje} \cos \varphi_e + d_{yje} \sin \varphi_e) \bar{i} + (-d_{xje} \sin \varphi_e + d_{yje} \cos \varphi_e) \bar{j}$$

$$\bar{r}_j = \bar{r}_e + \bar{r}_{je} = \begin{bmatrix} x_e + d_{xje} \cos \varphi_e + d_{yje} \sin \varphi_e \\ y_e + d_{xje} \sin \varphi_e + d_{yje} \cos \varphi_e \end{bmatrix}$$

$$\bar{v}_j = \dot{\bar{r}}_j = \begin{bmatrix} \dot{x}_e + \dot{\varphi}_e (-d_{xje} \sin \varphi_e + d_{yje} \cos \varphi_e) \\ \dot{y}_e + \dot{\varphi}_e (d_{xje} \cos \varphi_e + d_{yje} \sin \varphi_e) \end{bmatrix} = \begin{bmatrix} v_{jx} \\ v_{jy} \end{bmatrix}$$

$$v_j^2 = \langle \bar{v}_j, \bar{v}_j \rangle = v_{jx}^2 + v_{jy}^2 = \dot{x}_e^2 + 2\dot{x}_e \dot{\varphi}_e (-d_{xje} \sin \varphi_e + d_{yje} \cos \varphi_e) + \dot{\varphi}_e^2 (d_{xje}^2 + d_{yje}^2)$$

$$= \dot{x}_e^2 + \dot{y}_e^2 + \dot{\varphi}_e^2 (d_{xje}^2 \sin^2 \varphi_e + d_{xje}^2 \cos^2 \varphi_e + d_{yje}^2 \cos^2 \varphi_e + d_{yje}^2 \sin^2 \varphi_e - 2d_{xje} \sin \varphi_e d_{yje} \cos \varphi_e + 2d_{xje} \cos \varphi_e d_{yje} \sin \varphi_e) + 2\dot{\varphi}_e [\dot{x}_e (-d_{xje} \sin \varphi_e + d_{yje} \cos \varphi_e) + \dot{y}_e (-d_{xje} \cos \varphi_e - d_{yje} \sin \varphi_e)]$$

$$= \dot{x}_e^2 + \dot{y}_e^2 + \dot{\varphi}_e^2 (d_{xje}^2 + d_{yje}^2) + 2\dot{\varphi}_e [d_{xje} (-\dot{x}_e \sin \varphi_e - \dot{y}_e \cos \varphi_e) + d_{yje} (\dot{x}_e \cos \varphi_e - \dot{y}_e \sin \varphi_e)]$$

$$E_e = \sum_{j=1}^N E_j = \frac{1}{2} \sum_{j=1}^N m_j v^2 = \frac{1}{2} \left(\sum_{j=1}^N m_j \dot{x}_e^2 + \sum_{j=1}^N m_j \dot{y}_e^2 + \sum_{j=1}^N \dot{\varphi}_e^2 m_j (d_{xje}^2 + d_{yje}^2) + \sum_{j=1}^N m_j d_{xje} \cdot 2\dot{\varphi}_e (-\dot{x}_e \sin \varphi_e - \dot{y}_e \cos \varphi_e) + \sum_{j=1}^N m_j d_{yje} \cdot 2\dot{\varphi}_e (\dot{x}_e \cos \varphi_e - \dot{y}_e \sin \varphi_e) \right)$$

$$E_e = \sum_{j=1}^N m_j = M_e, \quad \sum_{j=1}^N m_j d_{xje} = S_y, \quad \sum_{j=1}^N m_j d_{yje} = S_x, \quad \sum_{j=1}^N m_j (d_{xje}^2 + d_{yje}^2) = J_{ze}$$

Ne aflăm în planul xOy , deci $S_x = S_y = 0$, deci $S_x = S_y = 0$

$$\Rightarrow E_e = \frac{1}{2} \dot{x}_e^2 M_e + \frac{1}{2} \dot{y}_e^2 M_e + \frac{1}{2} \dot{\varphi}_e^2 J_{ze}$$

În planul xOy , avem un singur moment de inerție plan: $\begin{cases} J_{xx}^0 \\ J_{yy}^0 \\ J_{xy}^0 \end{cases}$

8.7. Considerăm vectorul de coordonate generalizate:

$$q = \begin{bmatrix} x_1 \\ y_1 \\ \varphi_1 \\ \vdots \\ x_n \\ y_n \\ \varphi_n \end{bmatrix} \Rightarrow \dot{q} = \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{\varphi}_1 \\ \vdots \\ \dot{x}_n \\ \dot{y}_n \\ \dot{\varphi}_n \end{bmatrix}$$

Energia intrului sistem este:

$$E = \sum_{l=1}^n E_l = \frac{1}{2} \sum_{l=1}^n \dot{x}_l^2 M_l + \dot{y}_l^2 M_l + \dot{\varphi}_l^2 J_l$$

$$= \frac{1}{2} \cdot \dot{x}_1^2 M_1 + \dot{y}_1^2 M_1 + \dot{\varphi}_1^2 J_1 + \dot{x}_2^2 M_2 + \dot{y}_2^2 M_2 + \dot{\varphi}_2^2 J_2 + \dots + \dot{x}_n^2 M_n + \dot{y}_n^2 M_n + \dot{\varphi}_n^2 J_n$$

$$= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} \dot{q}_i \dot{q}_j \quad \text{este o formă pătratică în } \dot{q}, \text{ având}$$

matricea diagonală:

$$A = \begin{pmatrix} M_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & M_1 & 0 & 0 & \dots & 0 \\ 0 & 0 & J_1 & 0 & \dots & 0 \\ 0 & 0 & 0 & M_2 & \dots & 0 \\ 0 & 0 & 0 & 0 & M_2 & J_2 \\ 0 & 0 & 0 & 0 & 0 & \ddots \\ 0 & 0 & 0 & 0 & 0 & 0 & M_n \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & M_n \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & J_n \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & J_n \end{pmatrix}$$

$$E = \frac{1}{2} \dot{q}^T A \dot{q}$$

Ecuațiile lui Lagrange în cazul sistemelor cu variabile independente

Fie: $\begin{cases} f_1(q_1, \dots, q_n, t) = 0 \\ \vdots \\ f_p(q_1, \dots, q_n, t) = 0 \end{cases}$ legăturile omonome a sistemului

Notăm $F(q, t) = (f_1(q, t), f_2(q, t), \dots, f_p(q, t))$

unde $q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \Rightarrow$ Putem scrie legăturile ca: $F(q, t) = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$

Ecuațiile lui Lagrange au formă:

$$\frac{d}{dt} \left(\frac{\partial \tilde{T}}{\partial \dot{q}_k} \right) - \frac{\partial \tilde{T}}{\partial q_k} = Q_k + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_k}, \text{ unde } \lambda_j \text{ sunt}$$

multipliatorii lui
Lagrange

Calculăm Jacobianul lui F :

$$J_F = \begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \frac{\partial f_1}{\partial q_2} & \dots & \frac{\partial f_1}{\partial q_{3n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial q_1} & \frac{\partial f_p}{\partial q_2} & \dots & \frac{\partial f_p}{\partial q_{3n}} \end{bmatrix}$$

$$\text{Fie } \Delta = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_p \end{bmatrix}$$

Observăm că atunci când calculăm $J_F^T \Delta$, obținem:

$$\begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \dots & \frac{\partial f_p}{\partial q_1} \\ \frac{\partial f_1}{\partial q_2} & \dots & \frac{\partial f_p}{\partial q_2} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial q_{3n}} & \dots & \frac{\partial f_p}{\partial q_{3n}} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_p \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^p \lambda_j \cdot \frac{\partial f_j}{\partial q_1} \\ \sum_{j=1}^p \lambda_j \cdot \frac{\partial f_j}{\partial q_2} \\ \vdots \\ \sum_{j=1}^p \lambda_j \cdot \frac{\partial f_j}{\partial q_{3n}} \end{bmatrix}$$

Vectorul obținut are chiar termenii din ec. lui Lagrange.

Calculăm $\frac{\partial E}{\partial \dot{q}_k}$:

$$\begin{aligned}\frac{\partial E}{\partial \dot{q}_k} &= \frac{\partial}{\partial \dot{q}_k} \left(\frac{1}{2} \dot{q}^T A \dot{q} \right) = \frac{1}{2} \left[\frac{\partial \dot{q}^T}{\partial \dot{q}_k} A \dot{q} + \dot{q}^T \frac{\partial A}{\partial \dot{q}_k} \dot{q} + \dot{q}^T A \cdot \frac{\partial \dot{q}}{\partial \dot{q}_k} \right] = \\ &= \frac{1}{2} \left[(1, 0, 0, \dots, 1, \dots, 0) A \dot{q} + 0 + \dot{q}^T A \cdot \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right] \\ &= \frac{1}{2} [p_k^T A \dot{q} + \dot{q}^T A p_k]\end{aligned}$$

Daci notăm $p_k^T A \dot{q}$ cu B , atunci $\dot{q}^T A p_k = B^T$

$$= \frac{1}{2} (B + B^T) \quad \text{dimensiunile matricelor } p_k^T, A, \dot{q} \text{ sunt } 1 \times n \cdot n \times n \cdot n \times 1 = 1$$

$\Rightarrow B$ este un scalar

$$\Rightarrow \frac{\partial E}{\partial \dot{q}_k} = \frac{1}{2} \cdot 2 \cdot a_{k,k} \dot{q}_k = a_{k,k} \dot{q}_k$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial E}{\partial \dot{q}_k} \right) = a_{k,k} \ddot{q}_k$$

Rezultă:

$$\frac{d}{dt} \left(\frac{\partial E}{\partial \dot{q}} \right) = \frac{d}{dt} \begin{pmatrix} \frac{\partial E}{\partial \dot{q}_1} \\ \vdots \\ \frac{\partial E}{\partial \dot{q}_n} \end{pmatrix} = A \ddot{q}$$

$$\begin{aligned}\text{Calculăm } \frac{\partial E}{\partial q_k} &= \frac{1}{2} \left[\frac{\partial \dot{q}^T}{\partial q_k} A \dot{q} + \dot{q}^T \frac{\partial A}{\partial q_k} \dot{q} + \dot{q}^T A \frac{\partial \dot{q}}{\partial q_k} \right] \\ &= 0 + 0 + 0 = 0\end{aligned}$$

$$\text{Notăm } q = \begin{bmatrix} q_1 \\ \vdots \\ q_k \end{bmatrix}$$

Când scriem ecuațiile lui Lagrange obținem:

$$\left\{ \begin{array}{l} \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}_1} \right) - \frac{\partial \bar{E}}{\partial q_1} = Q_1 + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_1} \\ \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}_2} \right) - \frac{\partial \bar{E}}{\partial q_2} = Q_2 + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_2} \\ \vdots \\ \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}_{3n}} \right) - \frac{\partial \bar{E}}{\partial q_{3n}} = Q_{3n} + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_{3n}} \end{array} \right.$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}} \right) - 0 = Q + J_f^T \Delta$$

$$(-) \quad A \ddot{q} = Q + J_f^T \Delta$$

Când scriem $F = 0$

Al. un sistem cu n corpuri și p legături, adăugarea unei
 articulații va adăuga două componente funcției F , și înca
 doi multiplicatori Lagrange.

$$F = [f_1(q), \dots, f_p(q)] \rightarrow F = [f_1(q), \dots, f_p(q), f_{p+1}(q), f_{p+2}(q)]$$

$$f_{p+1} = x_{PA} - x_{PB} = x_A + l_{xA} \cos \varphi_A - l_{yA} \sin \varphi_A - (x_B + l_{xB} \cos \varphi_B - l_{yB} \sin \varphi_B)$$

$$f_{p+2} = y_{PA} - y_{PB} = y_A + l_{xA} \sin \varphi_A + l_{yA} \cos \varphi_A - (y_B + l_{xB} \sin \varphi_B + l_{yB} \cos \varphi_B)$$

$$\gamma_F = \begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \dots & \frac{\partial f_{p+1}}{\partial q_{3n}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_{p+2}}{\partial q_1} & \dots & \frac{\partial f_{p+2}}{\partial q_{3n}} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial y_1} & \frac{\partial f_1}{\partial \varphi_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial \varphi_n} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_{p+1}}{\partial x_1} & \dots & \frac{\partial f_{p+1}}{\partial x_A} & \frac{\partial f_{p+1}}{\partial y_A} & \frac{\partial f_{p+1}}{\partial \varphi_A} & \frac{\partial f_{p+1}}{\partial x_B} \\ \frac{\partial f_{p+2}}{\partial x_1} & \dots & \frac{\partial f_{p+2}}{\partial x_A} & \frac{\partial f_{p+2}}{\partial y_A} & \frac{\partial f_{p+2}}{\partial \varphi_A} & \dots \end{bmatrix}$$

Un sistem $q = \begin{bmatrix} q_1 \\ \vdots \\ q_{3n} \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \\ \varphi_1 \\ \vdots \\ x_n \\ y_n \\ \varphi_n \end{bmatrix}$

Calculăm derivatele parțiale ale funcțiilor f_{p+1} și f_{p+2}

$$\frac{\partial f_{p+1}}{\partial x_1} = 0, \frac{\partial f_{p+1}}{\partial x_2} = 0, \dots, \frac{\partial f_{p+1}}{\partial x_A} = 1, \frac{\partial f_{p+1}}{\partial y_A} = 0, \frac{\partial f_{p+1}}{\partial \varphi_A} = -l_{xA} \sin \varphi_A - l_{yA} \cos \varphi_A$$

$$\frac{\partial f_{p+2}}{\partial x_B} = -1, \frac{\partial f_{p+2}}{\partial y_B} = 0, \frac{\partial f_{p+2}}{\partial \varphi_B} = l_{xB} \sin \varphi_B + l_{yB} \cos \varphi_B, \dots, \frac{\partial f_{p+2}}{\partial \varphi_n} = 0$$

$$\frac{\partial f_{p+2}}{\partial q_k} = 0 \quad \forall q_k \in \{x_A, y_A, \varphi_A, x_B, y_B, \varphi_B\}$$

$$\frac{\partial f_{p+2}}{\partial x_A} = 0, \quad \frac{\partial f_{p+2}}{\partial y_A} = 1, \quad \frac{\partial f_{p+2}}{\partial \varphi_A} = l_{xA} \cos \varphi_A - l_{yA} \sin \varphi_A$$

$$\frac{\partial f_{p+2}}{\partial x_B} = 0, \quad \frac{\partial f_{p+2}}{\partial y_B} = -1, \quad \frac{\partial f_{p+2}}{\partial \varphi_B} = -l_{xB} \cos \varphi_B + l_{yB} \sin \varphi_B$$



Introducerea articulației adaugă următorii vectori la matricea J_F :

$$\begin{array}{l} \text{coord} \text{ lin A} \\ \left[\begin{array}{c} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ l_{xA} \sin \varphi_A - l_{yA} \cos \varphi_A \\ -l_{xA} \cos \varphi_A \\ 0 \\ \vdots \\ -1 \\ 0 \\ l_{xB} \sin \varphi_B + l_{yA} \cos \varphi_B \\ 0 \\ \vdots \\ 0 \end{array} \right]^T \end{array} \quad \begin{array}{l} \text{coord} \text{ lin A} \\ \text{coord} \text{ lin B} \end{array} \quad \begin{array}{l} \text{lin} \\ \left[\begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ l_{xA} \cos \varphi_A - l_{yA} \sin \varphi_A \\ 0 \\ -1 \\ -l_{xB} \cos \varphi_B + l_{yB} \sin \varphi_B \\ 0 \\ 0 \\ \vdots \\ 0 \end{array} \right]^T \end{array}$$

Adăugarea unei încastrări va adăuga componentele:

$$\left. \begin{aligned} f_{p+1} &= x_{PA} - x_{PB} \\ f_{p+2} &= y_{PA} - y_{PB} \end{aligned} \right\} \text{ la fel ca la articulație}$$

$$f_{p+3} = \varphi_B - \varphi_A - \varphi_0 = 0, \text{ unde } \varphi_0 \text{ este unghiul la momentul încastrării}$$

Derivatele parțiale ale lui f_{p+3} vor fi 0, respectiv:

$$\frac{\partial f_{p+3}}{\partial \varphi_B} = 1 \quad ; \quad \frac{\partial f_{p+3}}{\partial \varphi_A} = -1$$

$\Rightarrow$ Adăugarea încastrării va avea ca efect adăugarea vectorilor:

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ -l_{xA} \sin \varphi_A - l_{yA} \cos \varphi_0 \\ 0 \\ \vdots \\ 0 \\ -1 \\ 0 \\ l_{xA} \sin \varphi_A + l_{yA} \cos \varphi_0 \end{bmatrix}^T$$

$$1 \quad \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ l_{xA} \cos \varphi_A - l_{yA} \sin \varphi_A \\ 0 \\ \vdots \\ -1 \\ -l_{xB} \cos \varphi_B + l_{yB} \sin \varphi_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^T$$

coord.
leii A
|
coord.
leii B

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^T$$